Analytical Cumulative Examination
October 22, 1994

It has been proposed that an optical fiber can function as a dispersive device in a spectrometer, replacing the conventional prism or grating. In applications requiring a rugged configuration, the optical fiber-based system could have significant advantages. In this mode, one makes use of the fact that the refractive index of the core material varies with wavelength. For the spectrometer with the characteristics listed, answer the following questions:

Fiber Characteristics:

Optical fiber length; 1.0 kilometer
Refractive index (n) at 486 nm; 1.523
Refractive index (n) at 589 nm; 1.517
Pulse width of light entering fiber; 500×10^{-12} s
Photomultiplier tube time constant; 2.0×10^{-9} s
Optical fiber diameter; 0.10 mm
Optical attenuation: insignificant over 1.0 km

1. (05 points) What is the analogy to the slit in a conventional spectrometer?
2. (15 points) Calculate the time it would take pulses of light at both 486 nm and 589 nm to traverse the fiber.
3. (05 points) How does one scan the wavelength of such a spectrometer?
4. (20 points) Assuming that the refractive index of the core material changes linearly with wavelength, how long would it take to scan the wavelength range from 200-500 nm?
5. (25 points) What is the maximum spectral resolution that can be achieved with this optical fiber spectrometer ($R = \lambda/\Delta\lambda$)?
6. (05 points) For silica, the refractive index decreases with increasing wavelength over the range 200-1500 nm. Would you expect the resolution to be higher or lower in the near IR (1000 nm).
7. (05 points) What is the analogy to linear dispersion with the optical fiber system?
8. (20 points) An indicator is available that can be covalently linked to the end of the optical fiber. The molecular and ionic forms of the indicator have absorption profiles with maximum absorbances that differ by 30 nm. Explicitly describe how you would use the optical fiber spectrometer and this indicator to produce an optical fiber-based pH sensor. In your description include experimental detail concerning how the measurement would be made.